

EEGvis

A Bloomberg-style terminal for EEG. Every recording opens as one dense, linked workspace — waveform, 3D scalp field, spectra and ICA all driven by a shared time cursor — and it's a drop-in web UI for MNE-Python, launched from a notebook or the CLI.

These frames redraw the working backend (M0–M4) and reframe the front end as a tiled terminal. New elements are tagged **PROPOSED**; unresolved calls **OPEN Q**.



BUILT TODAY (REAL MNE, SMOKE-TESTED)

Upload → trace window → bandpass/notch/resample/re-ref/montage → ICA fit + topomap PNG + exclude/apply → Welch PSD + band power + band topomap → FIF export.

THIS DECK PROPOSES

Tiled docking workspace, shared time cursor across panels, react-three-fiber 3D scalp heatmap, live spectrogram, small-multiples, command bar + F-keys, provenance ledger, runnable-.py / report export, eegvis.launch(raw) plugin entry.

THREE RULES EVERY PANEL EARNS AGAINST

- 1 One recording, one screen**
 Absolute-black tiled grid. Waveform, 3D field, PSD, spectrogram, ICA visible at once and *linked* — hover a time in one, every panel snaps to it. No wizard, no page reloads.
- 2 Terminal-fast**
 GPU rendering (WebGL / react-three-fiber), windowed fetches, a : command bar and F-key shortcuts. A power user never touches the mouse; nothing blocks the read.
- 3 A plugin, not a silo**
 pip install eegvis; eegvis.launch(raw) from a notebook or eegvis file.edf from a shell. Every session exports the cleaned file *plus* the runnable MNE script that produced it.

TERMINAL AESTHETIC

Absolute-black (#000 / panel #0A0A0C), 1px seams, one blue accent (#4FB0FF) + one amber for alerts. Data set in a mono face; labels in the UI face. Density over whitespace: a panel shows numbers, not captions. Every panel has a 16px header strip (title · source · sync-lock · pop-out) and nothing else chrome-wise.

--bg #000

--panel

--seam

--accent

--alert

--fg

INPUT MODEL

SURFACE	DOES
: command bar	:hp 1 :lp 40 :notch 60 :ica 20 :goto 124.5s :export py
F1–F8	jump / toggle panels: F1 waveform · F2 3D · F3 PSD · F4 spectro · F5 ICA · F6 steps · F7 layout · F8 export
← → ↑ ↓	page window / change μ V-per-div; [] nudge cursor; b mark bad; a annotate
drag seams	resize tiles; drag header = re-dock; double-click header = solo

MOTION

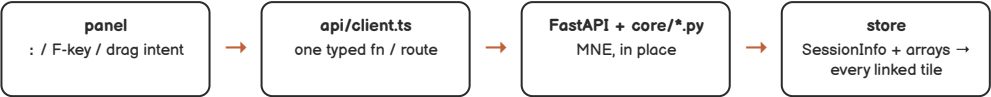
Framer Motion, 90–200ms, easing [0.22, 1, 0.36, 1]. The cursor line and panel data update every frame with no re-mount. Docking/solo animate; data never "loads" with a spinner — stale tiles dim to 60% until the fetch returns.

PLUGIN ENTRY & THE MNE BRIDGE

Nothing numeric in JS. The client sends intent; FastAPI mutates a live Raw/Epochs/ICA per session and returns a fresh summary + decimated arrays (or PNG). Three ways in, one session model:

ENTRY	MECHANISM
notebook	eegvis.launch(raw) → serializes to the running server (or spawns one) → opens a browser tab on that session PROPOSED
CLI	eegvis sub-04.edf → starts backend + frontend, opens the terminal PROPOSED
browser	drag a file onto the Connect screen NOW

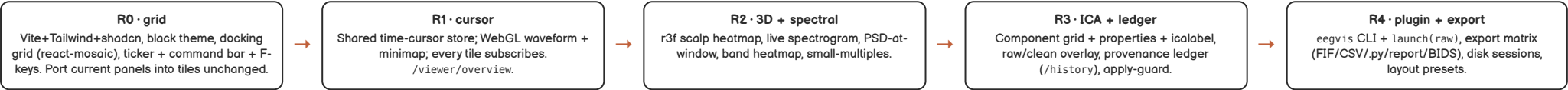
REQUEST PATH

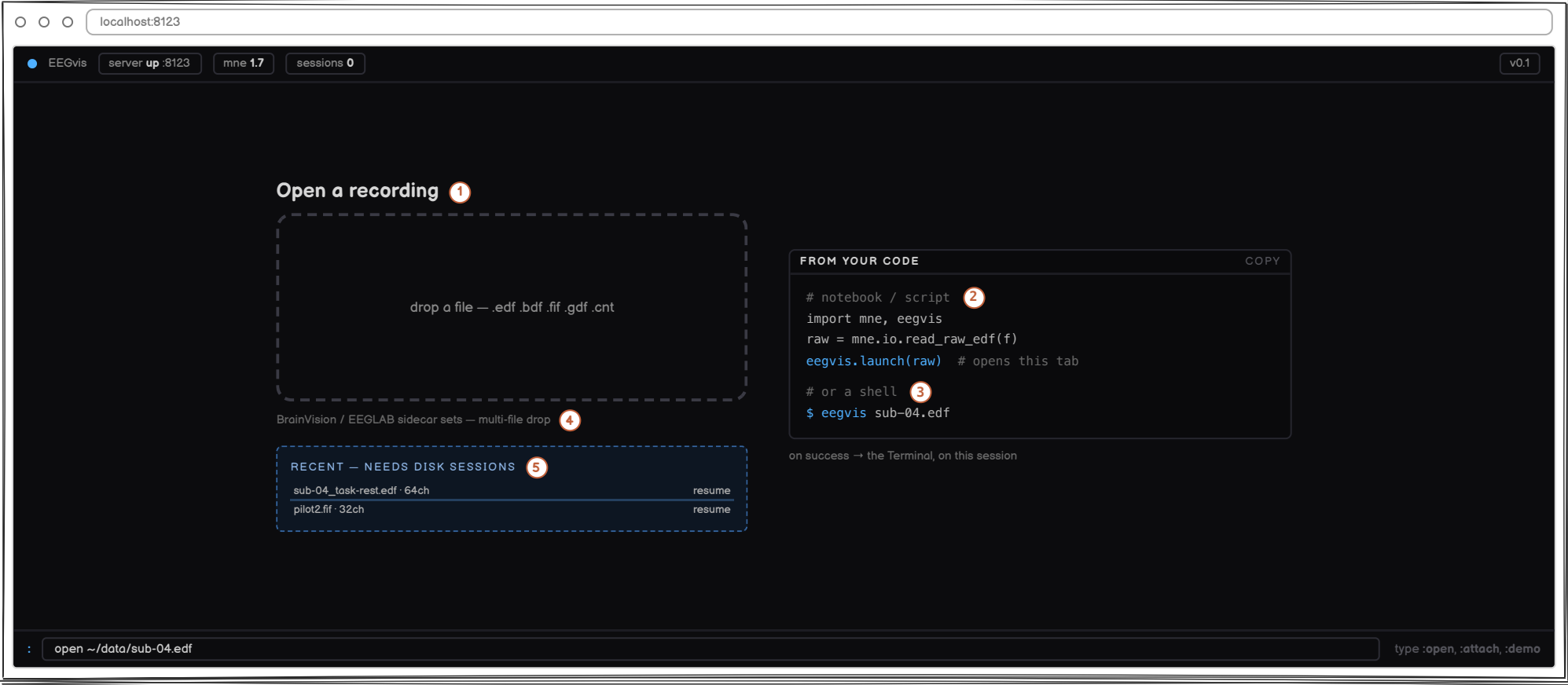


ENDPOINTS — LIVE VS PROPOSED

LIVE (WRAPPED)	PROPOSED (ADDITIVE)
/viewer/window	/viewer/overview — 1 val/s minimap
/preprocessing/*	/history — provenance ledger
/ica/{fit,topomap,exclude,apply}	/ica/{i}/properties · /icalabel
/spectral/{psd,band-power,band-topomap}	/spectral/tfr · /spectral/psd?window=
/export/{raw,epochs}.fif	/export/{summary.csv, pipeline.py, report.pdf, bids}
/sessions/*	/sessions/attach — from notebook object

BUILD ORDER — TERMINAL ON TOP OF THE WORKING BACKEND





BROWSER · THE ONLY NON-TILED SCREEN · CONNECT.TSX (WAS UPLOADPANEL)

NOTES & DECISIONS

Built: drag-drop / click upload of one self-contained file to /api/sessions/upload, error line, auto-route on success. This page proposes: the two code entry points, a recent list, multi-file import, a server-status ticker.

- 1

Dropzone. NOW the working path. Keep; add a parse-summary skeleton (ch / sfreq / duration / montage) before routing so bad files fail here.
- 2

eegvis.launch(raw). PROPOSED the headline feature. Python helper posts the in-memory Raw (fif buffer) to /sessions/attach, opens a browser tab. Makes it a real MNE plugin.
- 3

eegvis CLI. PROPOSED M5 packaging — one entry point boots backend+frontend and opens the terminal. Ships in R4.
- 4

Multi-file formats. OPEN Q BrainVision / EEGLAB need N files. One drop that groups by stem, or a guided "add the sidecar" step?
- 5

Recent sessions. PROPOSED blocked on disk persistence (M5). Until then an in-memory list only.
- 6

Command bar here. OPEN Q is :open / :demo worth wiring on the connect screen, or does the bar only appear once a session exists?

PRIOR ART

MNE's raw.plot() pops a Qt window from a notebook — local only, no export story. eegvis.launch(raw) is that reflex, but in a shareable browser session with a provenance trail.



BROWSER · TILED DOCKING GRID, SHARED TIME CURSOR, ALL PANELS LINKED · TERMINAL.TSX

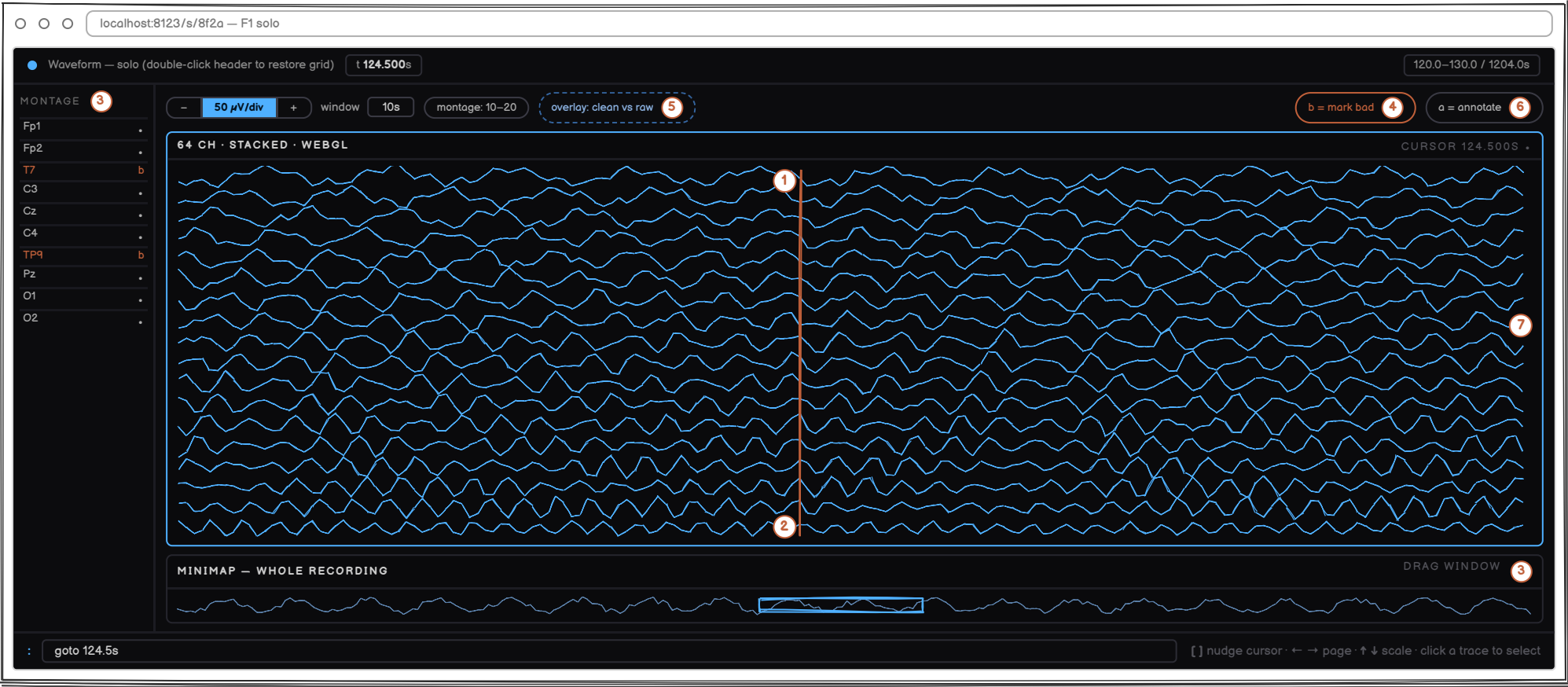
NOTES & DECISIONS

Built: the same data every tile shows — `/viewer/window`, `/spectral/*`, `/ica/*` — today rendered one panel at a time via step nav. **This page proposes:** showing them at once in a resizable grid, wired to one cursor.

- 1 Status ticker.** **[PROPOSED]** the live `SessionInfo` as a single scannable strip; updates optimistically after every mutation. Amber segment = something needs attention.
- 2 Shared time cursor.** **[PROPOSED]** the core idea. Hover `/ :goto` sets `t` in a store; waveform playhead, 3D field, PSD window and spectrogram marker all read it. One `/viewer/window` + slices drive the rest.
- 3 3D scalp field.** **[PROPOSED]** react-three-fiber head mesh, per-vertex colour interpolated from sensor values at `t` (or band power). Replaces / augments MNE's static topomap PNG. **[OPEN Q]** ship a generic head or per-montage?
- 4 PSD @ window.** **[PROPOSED]** Welch on the *visible* segment, not the whole file — needs `/spectral/psd?start&dur`. Recomputes (debounced) as you scrub.
- 5 Spectrogram.** **[PROPOSED]** continuous Morlet/multitaper on one channel, full-recording, cursor marker synced. New `/spectral/tfr`. Channel picked from the header.
- 6 Docking + F-keys + .** **[OPEN Q]** library: react-mosaic vs dockview vs golden-layout. Persist layout per session? Command grammar frozen now so muscle memory holds.

PRIOR ART

Bloomberg Terminal (linked panels, command line, F-keys), Persyst & Natus review stations (clinical EEG density), TradingView (docked charts). None connect a 3D field + spectra + ICA to one scrub. That link is the product.



BROWSER · THE SPINE — ITS CURSOR + SELECTION DRIVE EVERY OTHER TILE · WAVEFORMPANEL.TSX (WAS CHANNELVIEWER)

NOTES & DECISIONS

Built: Plotly scattergl stacked traces at a fixed 100 μ V offset, prev/next window, numeric window length, μ V hover readout, fetch of /viewer/window (start, duration, channels, max_points). This page proposes: WebGL renderer, the cursor, inline bad-marking, annotations, a clean/raw overlay.

- 1

Render. PROPOSED regl/WebGL line layer — 60fps pan, 256-ch HD-EEG without choking. Plotly stays as a fallback flag through R0.
- 2

Time cursor. PROPOSED a vertical line that follows the mouse (or :goto), published to the shared store. This is what makes the 3D field / PSD / spectrogram "live".
- 3

Minimap. PROPOSED downsampled whole-recording strip; drag the window box or click to jump. New read-only /viewer/overview (~1 value/s/channel, or RMS).
- 4

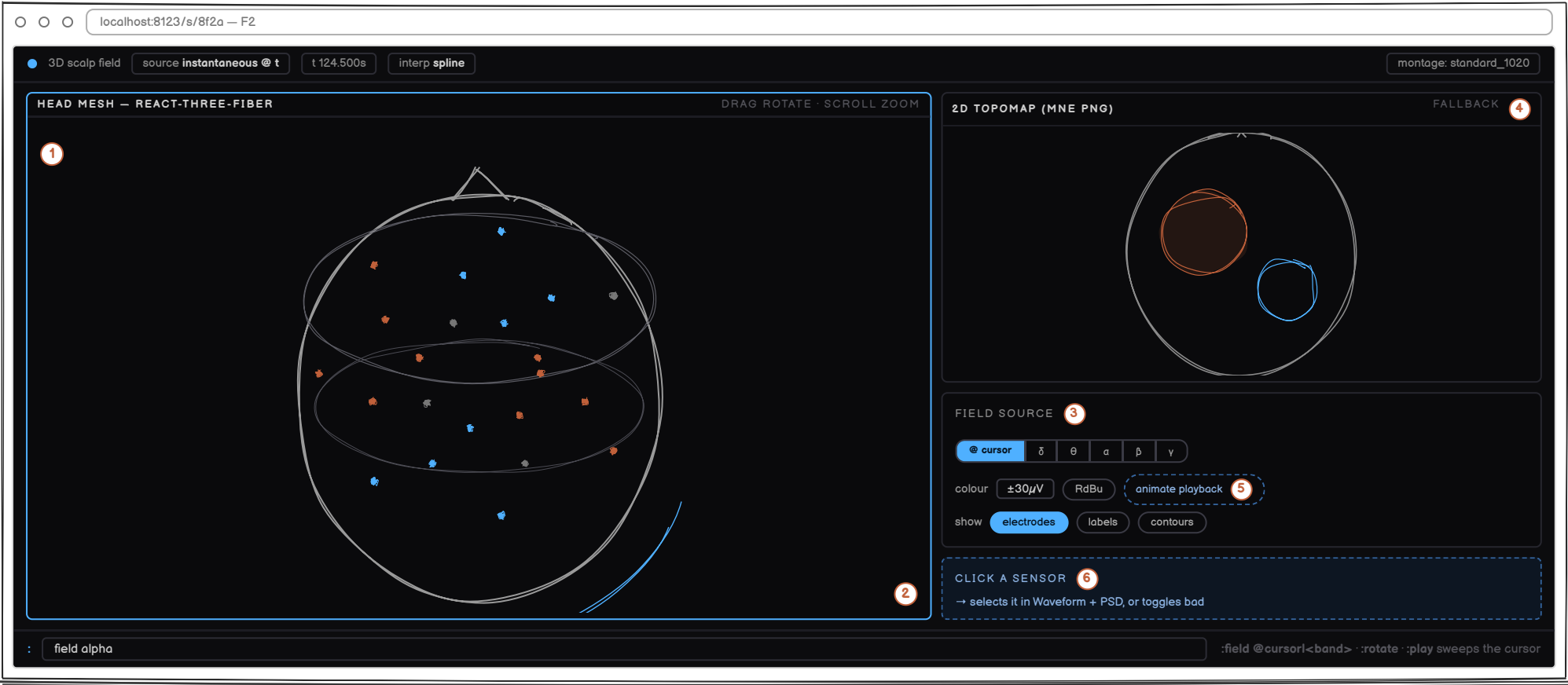
Mark bad (b). PROPOSED click a trace/name or press b on the selected channel → POST /preprocessing/bad-channels full set; row + trace go amber instantly, ticker updates.
- 5

Clean-vs-raw overlay. PROPOSED after ICA-apply / filtering, ghost the pre-step signal behind the current one so an edit's effect is visible in situ. Needs the original retained (see ledger).
- 6

Annotations (a). OPEN Q write to raw.annotations; these later seed epoching. Free-text label vs a fixed vocabulary? Drag to set duration.

PRIOR ART

mne-qt-browser is the interaction bar: buttery pan, click-to-mark-bad, spacebar annotate. We match it in the browser, add the minimap and the cross-panel cursor it has no concept of.



BROWSER · LIVE SCALP INTERPOLATION, CURSOR-DRIVEN · SCALPFIELD.TSX + REUSES /SPECTRAL/BAND-TOPOMAP

NOTES & DECISIONS

Built: server-rendered topomap PNGs from `mne.viz.plot_topomap` — one per ICA component, one per frequency band, fetched on demand. This page proposes: an interactive 3D field in the client, cursor/band-driven, with the PNG kept as a trusted fallback.

- 1

Head mesh. PROPOSED r3f + a low-poly scalp; per-vertex colour from sensor positions via RBF / spherical-spline interpolation in a shader or a small WASM pass. OPEN Q interpolate client-side, or ask the backend for a value grid per t?
- 2

Rotate / zoom / camera presets. PROPOSED orbit controls + top/left/right/back buttons. Camera state per workspace.
- 3

Field source. PROPOSED toggle: instantaneous voltage @ cursor, or band power ($\delta \dots \gamma$) over the window. Band mode reuses /spectral/band-power.
- 4

2D fallback. NOW keep the MNE PNG topomap in a sub-tile — it's the reference render and needs zero new code. 3D is the "wow", 2D is the check.
- 5

Playback. PROPOSED :play sweeps the cursor at Nx speed; the field animates like an evoked-response movie. Big demo value, needs smooth prefetch of the window.
- 6

Sensor picking. PROPOSED raycast to nearest electrode → select in Waveform/PSD or mark bad. Ties the 3D view into the same selection store as everything else.

PRIOR ART

MNE's `plot_topomap` / Brain (PyVista) 3D, EEGLAB `headplot`, Fieldtrip layouts. All are static images or a separate desktop GL window. Putting it in the terminal, cursor-linked, is new.



BROWSER · FOUR LINKED SPECTRAL VIEWS · SPECTRALWORKSPACE.TSX (WAS SPECTRALPANEL)

NOTES & DECISIONS

Built: a "Compute PSD (1-45 Hz)" button → /spectral/psd (Welch) + /spectral/band-power, a Plotly line-per-channel PSD, then band buttons that each fetch a topomap PNG. This page proposes: aggregate PSD, a heatmap, a real spectrogram, small multiples, a source switch.

- 1

Compute trigger. NOW explicit button (PSD isn't cheap). Keep, plus auto-recompute (debounced) when bound to the cursor window; show "computed Ns ago / reflects step 6".
- 2

PSD plot. PROPOSED default to mean ± IQR band (line-per-channel is noise past ~32 ch); individual channels on hover / from selection.
- 3

Band heatmap. PROPOSED /band-power already returns {band: [per-channel]} — render channelxband directly; click a cell → that band on the 3D field + that channel selected.
- 4

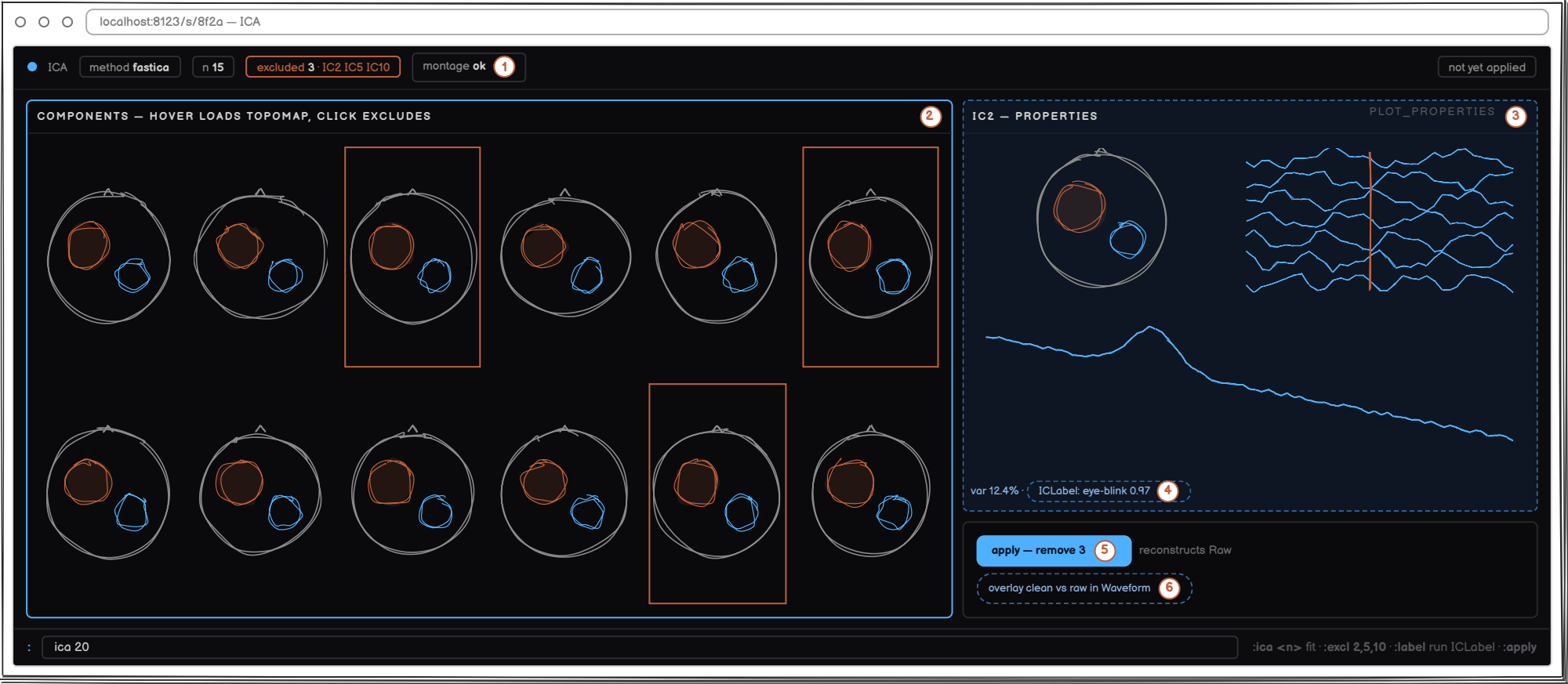
Spectrogram / TFR. PROPOSED new /spectral/tfr (Morlet or multitaper) on one channel, continuous over the recording, cursor marker synced with the waveform.
- 5

Small multiples. PROPOSED a 8x8 grid of thumbnail PSDs laid out by montage position — spot a bad channel or an occipital alpha bump at a glance.
- 6

Raw vs Epochs. OPEN Q same math, different object. Fold a lightweight epoching control into this screen, or keep a separate Epochs step feeding it?

PRIOR ART

MNE Spectrum.plot / compute_tfr, FieldTrip ft_multiplotER, Persyst spectrogram (clinical trend). Our bet: the montage-shaped small-multiples grid as the default read, topomap/heatmap for the summary.



BROWSER · GRID + PROPERTIES + APPLY, LINKED TO WAVEFORM OVERLAY · ICAWORKSPACE.TSX

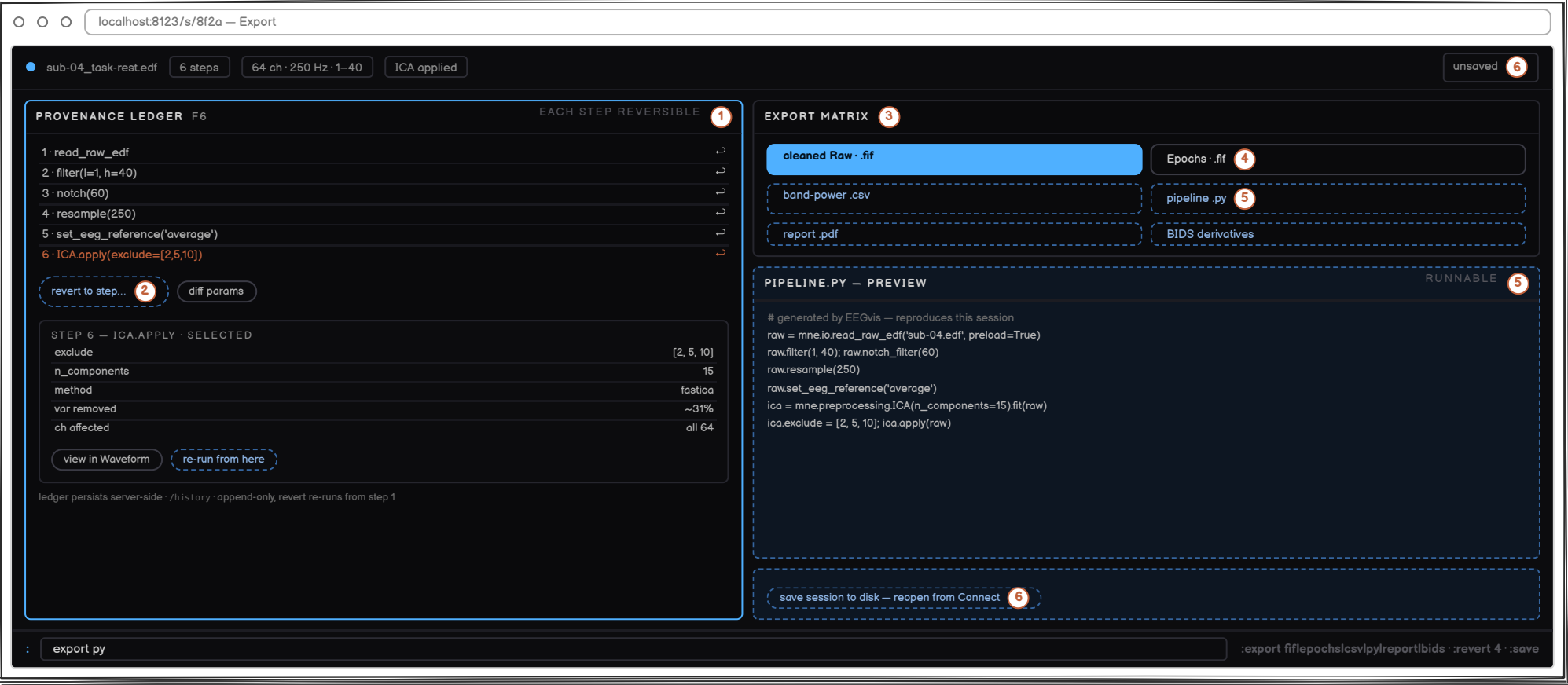
NOTES & DECISIONS

Built: `n_components` input + Fit (`fastica`), a grid showing `ICn · variance%`, lazy topomap PNG on hover (`/ica/components/{i}/topomap`), click toggles exclude (`/ica/exclude` full list), Apply (`/ica/apply`). Excluded cards go amber. This page proposes: a properties tile, ICLabel, an apply guard, the clean/raw overlay.

- 1 **Montage gate.** `NOW` topomaps are empty with no montage — today a hint only. Make Fit a hard precondition; ticker shows montage state, links to fix it.
- 2 **Grid load.** `OPEN Q` hover-lazy hides everything until touched. Eager-load all N topomaps after fit (N is small), or keep lazy with "load all"?
- 3 **Properties tile.** `PROPOSED` click a component → topomap + IC time course + IC spectrum + variance (MNE `plot_properties`). New `/ica/components/{i}/properties`.
- 4 **ICLabel.** `PROPOSED` wrap `mne-icalabel` → class + probability per IC (brain / eye / muscle / heart / line / channel). Badge on each card; user still confirms.
- 5 **Apply is one-way.** `OPEN Q` `/ica/apply` rewrites the Raw irreversibly. Gate behind a Dialog, record as a ledger step so revert re-runs the pipeline from the original.
- 6 **Overlay compare.** `PROPOSED` before applying, ghost the reconstructed-without-excluded signal behind the current one in the Waveform tile — see exactly what you're removing.

PRIOR ART

MNE `ICA.plot_components + plot_properties`; ICLabel (EEGLAB) is the classifier people trust. We fold grid + properties + label + apply-preview into one tile group instead of four matplotlib windows.



NOTES & DECISIONS

Built: two links hitting /export/raw.fif and /export/epochs.fif that stream the mutated MNE object, with a one-liner that downloads reflect every step so far. This page proposes: the ledger, revert, and five more export targets — chief among them runnable Python.

- 1

Provenance ledger. PROPOSED every mutating call appends {op, params, ts} server-side (/history). It's the audit trail, the summary, and the source for the .py export — one structure, many outputs.
- 2

Revert. OPEN Q in-place mutation has no undo. Re-run the recorded ledger from the original Raw up to step N (cheap-ish, deterministic) vs snapshot per step (fast, memory-heavy). Lean re-run.
- 3

Export matrix. NOW FIF works. PROPOSED CSV (band power + exclusion log — already scoped in the arch doc), report PDF (topomaps + PSD + step list), BIDS derivatives.
- 4

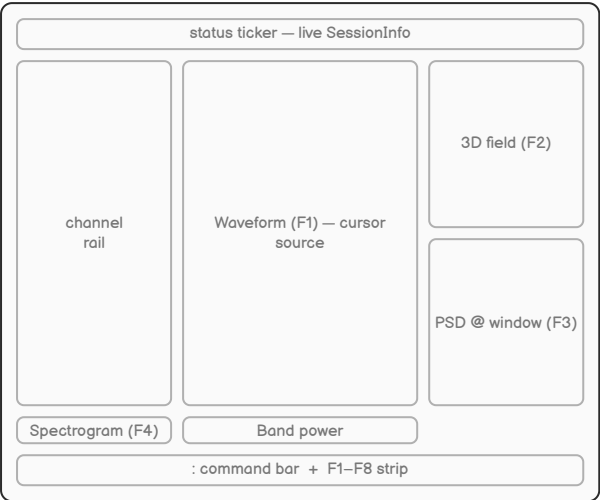
Epochs. NOW button 404s unless an epoching route ran. Disable until epochs exist; link to the control.
- 5

pipeline.py. PROPOSED render the ledger as an mne script. This is what makes EEGvis defensible: a GUI that emits reproducible code, so the terminal is a starting point, not a lock-in.
- 6

Persistence. OPEN Q M5. Where do sessions live (working dir / XDG data), TTL for abandoned ones, and does :save write the ledger + a fif or a full session bundle?

PRIOR ART
EEGLAB's EEG.history + pop_saveset, MNE-BIDS write_raw_bids, nipytype provenance. We converge them: cleaned file + audit trail + runnable script, in one panel.

DEFAULT DOCK — "TERMINAL" WORKSPACE



DRAG SEAMS TO RESIZE · DRAG HEADER TO RE-DOCK · DBL-CLICK = SOLO · LAYOUT SAVED PER SESSION

WORKSPACE PRESETS PROPOSED — SAME SESSION, DIFFERENT TILE SET

Terminal

Everything linked. The default read: waveform + 3D + spectra + bands.

Clean

Waveform big + ICA grid + ledger. For getting artefacts out fast.

Report

3D field + PSD + band heatmap + step list, export-framed for a screenshot / PDF.

GLOBAL SURFACES

- Command bar.** every mutating route has a verb; Tab-completes; echoes into the ledger.
- Palette (%K).** fuzzy over commands + channels + components + steps.
- F1–F8.** focus / solo a panel; Esc restores the dock. Muscle memory across sessions.
- Shared cursor + selection.** one store; every tile subscribes. The spine of the whole UI.

CROSS-PANEL FLOWS

- Connect → Terminal.** upload / launch(raw) / eegvis file → session → docked terminal.
- Waveform cursor → all.** hover / :goto sets t; 3D field, PSD, spectrogram redraw at t.
- 3D sensor ↔ Waveform / PSD.** click an electrode → select that channel everywhere, or mark it bad.
- Band heatmap cell → 3D + Waveform.** click → that band on the field, that channel selected.
- ICA apply → Waveform overlay + ledger.** ghosted clean/raw compare; step appended, revertible.
- Ledger → Export.** the step list is the summary, the .py, the report body.

NOT PRESENT: AUTH · ONBOARDING · MULTI-USER · REAL-TIME/STREAMING (LSL) · SOURCE LOCALIZATION / MRI — EXPLICIT V1 NON-GOALS